[6] TPS61086 - Boost 3.3V to 6V

COMP (pin 1): error amplifier compensation pin; connect the Rcomp + Ccomp network to AGND to stabilize the loop (do not route it near SW).

FB (pin 2): feedback input; connect it to the R1/R2 resistor divider from VOUT (after the diode) to set the output voltage (e.g. 6 V).

EN (pin 3): enable / shutdown; high level = active, low level = off. Often tied to IN for "always ON", or driven by a GPIO.

AGND (pin 4): analog ground (reference for sensitive signals: FB/COMP/SS/MODE). Keep it "clean" and connect it to the GND plane in a controlled way.

PGND (pin 5): power ground (return path for switching currents). Connect it to the GND plane with a short/wide copper area close to the power capacitors.

SW (pins 6-7): switch node; connect it to L and the Schottky diode. This is the noisiest node → keep the trace short, minimize the loop area, and keep it away from FB/COMP.

IN (pin 8): power input (2.3-6 V); decouple it as close as possible with Cby ~1 µF + Cin (e.g. 10 µF) to GND.

MODE (pin 9): mode selection; MODE = High → Forced PWM (more stable frequency, better for noise), MODE = Low → PFM (better efficiency at light load).

SS (pin 10): soft-start; add a capacitor (Css) to limit inrush current at startup. Left open = no soft-start.

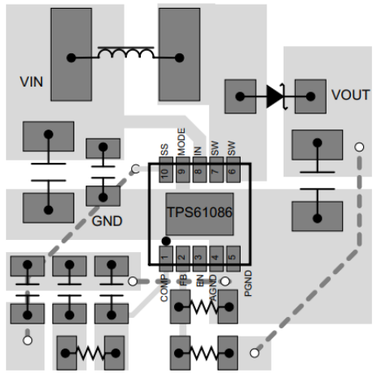
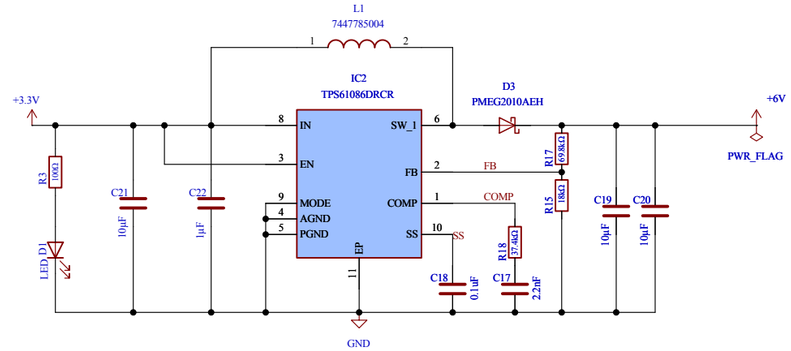


Figure 26. TPS61086 Layout Example

Pin usage and layout example



Boost 3.3 V -> 6.04 V

About

The TPS61086 device is a high-frequency, high-efficiency DC-to-DC converter with an integrated 2.0-A, 0.13-Ω power switch capable of providing an output voltage up to 18.5 V. The implemented boost converter is based on a fixed frequency of 1.2-MHz, pulse-width-modulation (PWM) controller that allows the use of small external inductors and capacitors and provides fast transient response.

At light-load, the device can operate in Power Save Mode with pulse-frequency-modulation (PFM) to improve the efficiency while keeping a low-output voltage ripple. For very noise-sensitive applications, the device can be forced to PWM Mode operation over the entire load range by pulling the MODE pin high. The external compensation allows optimizing the application for specific conditions. A capacitor connected to the soft-start pin minimizes inrush current at start-up.



Test points

Comments: **References:** TI TPS61086 datasheet; L1 7447785004, D3 PMEG2010AEH	Company: **OPEN THRUST LAB**		Variant:	Git Hash: 349b121
	Board Name: **AD7190**		Project Name: **OPEN_WEIGHT**	
Sheet Title: TPS61086 - Boost 3.3V to 6V	File Name: TPS61086 - Boost 3V3 to 6V.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/TPS61086_3.3V_to_6V/		Reviewer:	Size: **A3**	Sheet: **6** of **10**